

Project Proposal: Predicting Customer Conversion in Digital Marketing

Problem statement formation

Which customers are most likely to convert from a digital marketing campaign, and which demographic, behavioral, and campaign-level factors most strongly drive that conversion? The goal is a model that, given a customer's profile and the campaign targeting them, predicts conversion and surfaces the levers marketers can actually pull.

Context

Marketing budgets face constant ROI scrutiny, and a large share of spend is wasted on customers who were never likely to convert. A conversion model lets teams concentrate spend on high-probability segments, match customers to the channels and campaign types they respond to, and justify budget decisions with evidence rather than intuition. This dataset captures exactly the customer-and-campaign signals—demographics, ad spend, multi-touch engagement, and loyalty history—that marketing teams use to make those targeting calls.

Criteria for success

Build a model that predicts Conversion reliably while properly handling the dataset's heavy class imbalance, since the data skews heavily toward converted customers (roughly 88% positive), meaning the non-converters are the minority and the harder, more valuable class to identify. Beyond raw accuracy, the model should surface the features that most strongly drive conversion, producing actionable insights for channel and campaign strategy, all while guarding against overfitting and leakage.

Scope of solution space

Establish a model that effectively manages class imbalance, likely through the establishment of a log regression baseline and then exploration of tree-based or gradient boosting models for further improvement.

Constraints

The data is synthetic, so relationships may be cleaner and more separable than real-world campaigns, and generalization to live data is unproven. The class imbalance means accuracy is misleading and minority-class (non-converter) recall is the real challenge. There is a leakage risk worth investigating early: ConversionRate and possibly ClickThroughRate may be near-deterministic proxies for the target and could need to be dropped or carefully handled before modeling. Finally, two columns (AdvertisingPlatform and AdvertisingTool) are redacted, and the dataset carries no timeframe, which limits temporal and platform-level analysis.

Stakeholders

Marketing campaign managers and demand-generation teams deciding where to focus spend; media buyers and channel owners optimizing targeting; marketing analytics and operations teams; and finance or leadership stakeholders accountable for marketing ROI.

Data sources

Ryan Lange
June 2026

[Kaggle — Predict Conversion in Digital Marketing Dataset \(Rabie El Kharoua\)](#). A single synthetic CSV of 8,000 customer records across 20 columns spanning demographics, campaign attributes, engagement metrics, and loyalty indicators, with a binary Conversion target.